12 The Linkage Room

Two loci at once

CORE PATH · BSC	EXTENSION · MSC	BEFORE THIS ROOM
about 35 min	**+ 30 min**	The Hardy–Weinberg Room (11). The Selection Room (05) helps for Part 2.
steps 1–4	steps 5–7	

OPEN THE ROOM

`https://evolinus.github.io/zooe/teaching/evolutionary-laboratories/#linkage`

Two loci on the same chromosome, a distance r apart. Part 1 builds an association between A1 and B1 and lets recombination pull it apart. Part 2 drops two beneficial mutations into different individuals and counts how often the population manages to keep both. This guide asks how long associations last and what they cost.

LEARNING OBJECTIVES

After the core path you should be able to

- Define linkage disequilibrium D and read it from the gamete square.
- Predict how quickly an association decays at different recombination rates.
- Contrast Hardy–Weinberg (reached in one generation) with linkage equilibrium (only approached).
- Explain Hill–Robertson interference: why two good mutations on different chromosomes compete.

The extension adds

- Use $D_t = D_0(1 - r)^t$ and $t_{½} \approx 0.69/r$ to predict decay.
- Use $2Nrf_0^2$ to predict how fast the double mutant is made, and how that depends on N.
- Connect $r = 0$ to asexual reproduction and the advantage of sex.

SUGGESTED ROUTES

- **40-minute practical (BSc):** steps 1–4 and questions 1–3.
- **Sex and recombination seminar:** the Reproduction Room core path, then steps 3 and 7 here.
- **MSc:** steps 5–7 and questions 4–6 (about 30 minutes).

How to work through it. Every step asks you to *predict* before you press anything, and to *explain* afterwards. Write the prediction down and commit to it: a wrong prediction you can account for teaches more than a right one you guessed. Runs are random, so where a step says so, run it more than once before you decide what happened.

1 Free recombination

8 min

SET UP

Part 1. Recombination r = **0.5**, D_0/D_{max} = 1, p = 0.5.

PREDICT FIRST

At r = 0.5 the loci behave as if on different chromosomes. How many generations until most of the association is gone?

RUN

Press **Auto-play**.

LOOK AT

the gamete square, the decay curve, and the D readout.

EXPLAIN

What does r = 0.5 mean physically, and why does the association halve every generation?

2 Tight linkage

10 min

SET UP

r = **0.005**. Press **Reset to generation 0**.

PREDICT FIRST

After 60 generations, what fraction of the starting association will be left?

RUN

Auto-play for about 60 generations.

EXPLAIN

Why does tight linkage let an association persist? Name a real situation where alleles at nearby loci are found together far more often than chance would predict.

3 Two good mutations

12 min

SET UP

Part 2: $N = 50$, $s = 0.20$, starting frequency 0.050, 200 replicates. Use $r = 0.010$ from Part 1.

PREDICT FIRST

Each mutation on its own would usually spread. In what share of runs will the population keep *both*? Will that share be higher or lower than with free recombination?

RUN

Press **Watch one population**, then **Run all replicates**.

LOOK AT

the bars in *How often both mutations survive*, at your r against free recombination.

EXPLAIN

Why are the two mutations rivals rather than allies until recombination puts them on one chromosome?

4 One generation, or never quite

5 min

SET UP

Nothing to change.

PREDICT FIRST

Hardy–Weinberg proportions are reached in one generation. Is linkage equilibrium?

EXPLAIN

Explain the difference, in terms of what random mating can and cannot reshuffle in one generation.

EXTENSION · MSC

5 Half-lives

10 min

SET UP

Part 1: $r = 0.01$, 0.05 and 0.20 in turn.

PREDICT FIRST

Calculate $t_{1/2} = \ln 2/[-\ln(1 - r)]$ for each.

RUN

Auto-play from generation 0 and note when D falls to half of D_0.

EXPLAIN

Why is the approximation $0.69/r$ good at small r and poor at large r?

6 Population size and interference

12 min

SET UP

Part 2, $r = 0.01$. $N = 50$, then $N =$ **500**.

PREDICT FIRST

The expected number of double-mutant chromosomes made per generation is about $2Nrf_0^2$. Calculate it at both sizes. Will interference be stronger or weaker in the larger population?

RUN

Run all replicates at each N.

EXPLAIN

Why does a larger population get both mutations onto one chromosome more often? What does this say about adaptation in small populations?

7 No recombination at all

8 min

SET UP

Part 2 with $r =$ **0**.

PREDICT FIRST

What share of runs will keep both mutations?

RUN

Run all replicates.

EXPLAIN

An asexual lineage lives at $r = 0$ across its whole genome. Use this result to state one advantage of sex, and connect it to the Reproduction Room.

DISCUSSION AND WORKSHEET QUESTIONS

1. BSc Why might two alleles be found together more often than expected in a population without any selection favouring the combination?
2. BSc A disease allele is in strong linkage disequilibrium with a marker. Why is that useful to a geneticist?
3. BSc Explain in one paragraph why the one-locus rooms are a simplification.
4. MSc Genetic hitchhiking: how can a neutral allele rise to high frequency because of selection at a linked locus? Use D in your answer.
5. MSc Why do regions of low recombination in many genomes carry less neutral diversity?
6. MSc The room uses multiplicative fitness, so there is no epistasis. How would positive or negative epistasis between A_1 and B_1 change Part 2?

CHECKING YOUR REASONING

Read this after you have written your own answers. These are what the model should show on most runs — not the only acceptable explanations.

- **Step 1.** The association halves each generation: after five generations about 3% is left. $r = 0.5$ means a recombinant gamete is as likely as a parental one.
- **Step 2.** About $(1 - 0.005)^{60} \approx 0.74$ of the association remains. Nearby alleles stay together for many generations — haplotype blocks in human genomes, for example.
- **Step 3.** Both are kept noticeably less often at $r = 0.01$ than with free recombination; expect roughly half the runs against about seven in ten, give or take a few points from one batch of replicates to the next.
- **Step 4.** Random mating rebuilds genotypes from allele frequencies at once, but an association between loci is only broken by recombination events, a fraction r per generation.
- **Step 5.** About 69, 13.5 and 3.1 generations. At large r, $-\ln(1 - r)$ is no longer close to r.
- **Step 6.** About 0.0025 copies per generation at $N = 50$ and 0.025 at $N = 500$: the larger population builds the double mutant ten times as fast, so it keeps both more often.
- **Step 7.** None: with no recombination the double mutant can never be made, and one of the two mutations is always lost.